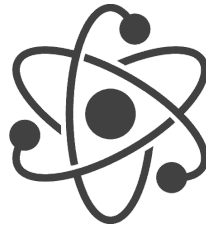


Using Volume EM Containers

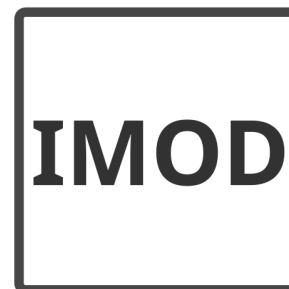
An Introduction with DAaaS Training



Are these slides for you?

- Looking to get started with volume EM software:

Napari, Fiji, IMOD, MIB



- Little-to-no computing experience
- Recommended: Access to DAaaS Training Workspace



What are Containers?

Containers are a software bundle that includes everything a computer program needs to run.

Think of a physical CD  with a preinstalled application: you can insert it into any computer and run the program immediately—no tricky installation required!



RFI Volume EM Containers

As a [CCP-VolumeEM](#) initiative, the Rosalind Franklin Institute (RFI) has created container 'CDs' for:

- 1.** Napari with CLEM-Reg and Empanada Plugins
- 2.** Fiji with MoBIE
- 3.** IMOD
- 4.** MIB2

The following slides give an overview of **1–4**. Further details and links to additional learning materials are in [our documentation](#).

RFI Volume EM Containers (2)

The aim of the container initiative is to increase accessibility of volume EM software.

 DAaaS users: the containers are already loaded on the training workspace, go to `Applications > Software` (bottom-left of screen).

Want to use the containers elsewhere? We suggest asking your computing team to help get up and running using Apptainer or Docker.

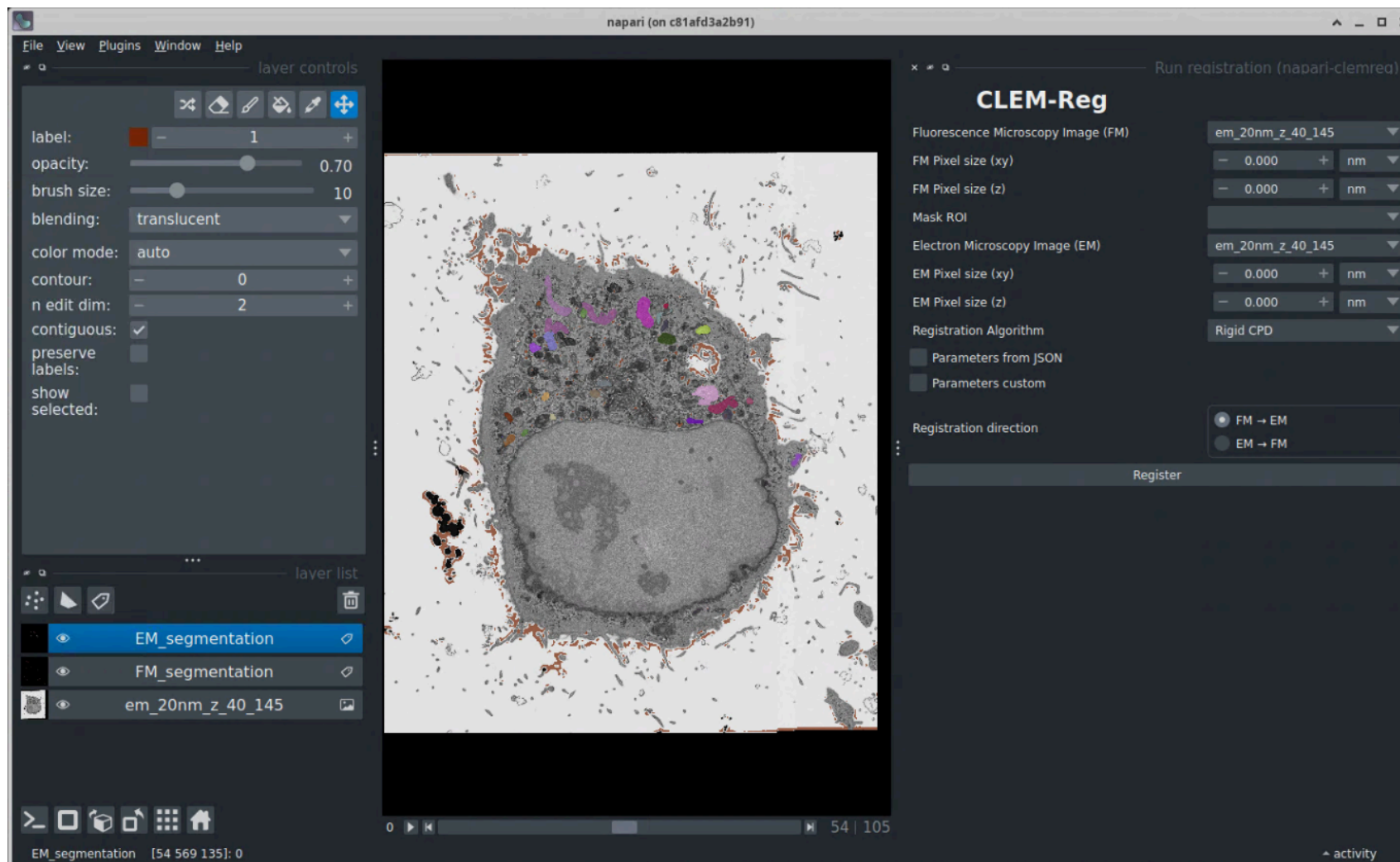
Napari with CLEM-Reg and Empanada

Napari is a modern interactive viewer with plugins for automated registration and segmentation in EM.

Use for: automated point cloud based registration ([CLEM-Reg](#)) and deep-learning segmentation of 2D/3D cellular structures ([Empanada](#)).



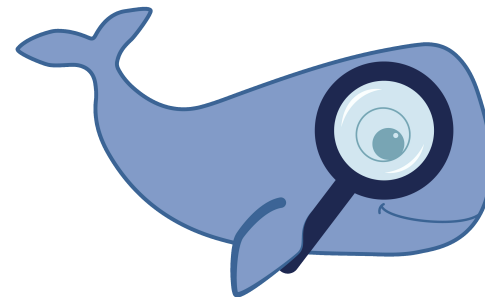
What Napari looks like (CLEM-Reg)



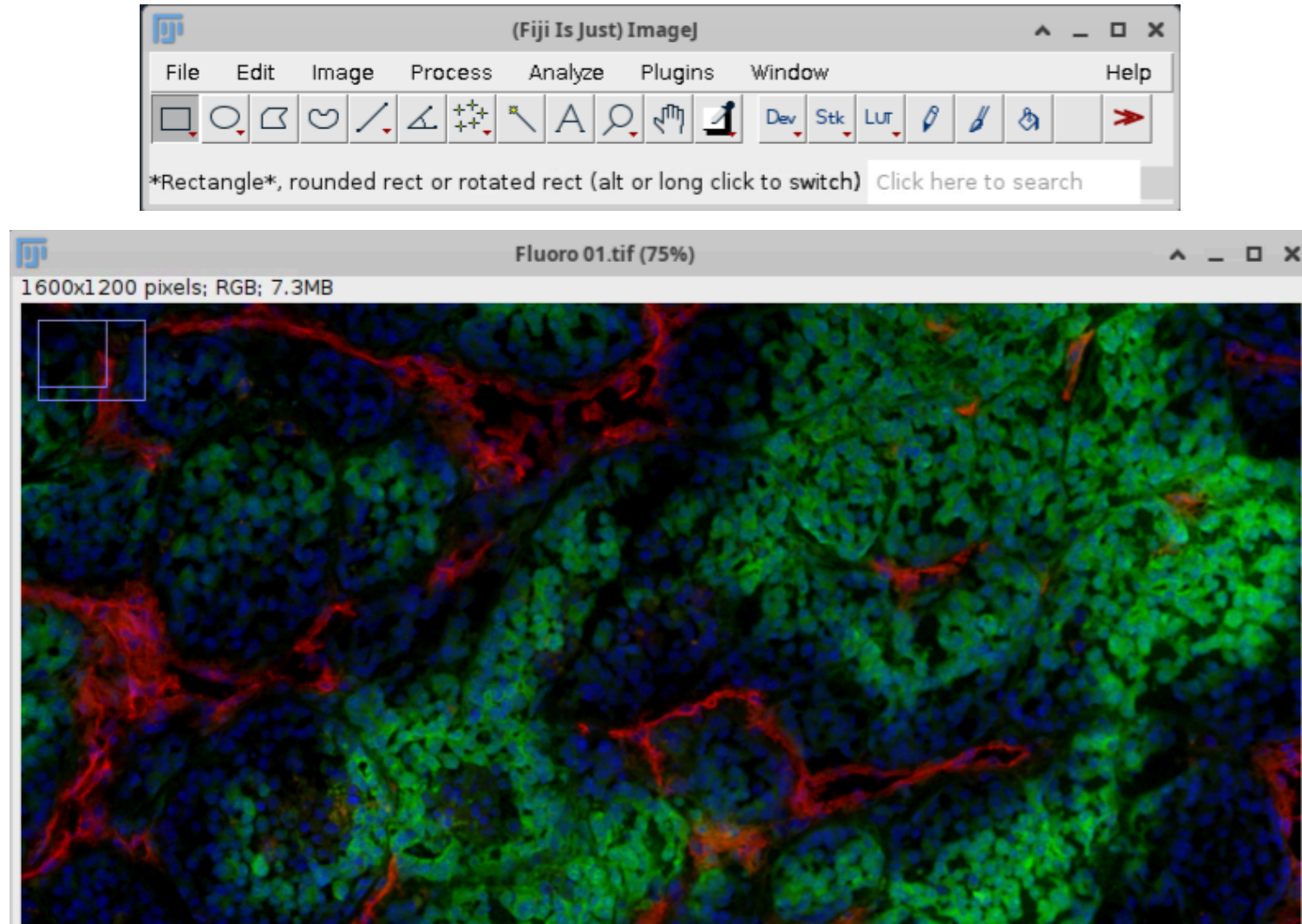
Fiji with MoBIE

[Fiji](#) is a widely-used image processing application that uses the [ImageJ](#) platform. The [MoBIE](#) plugin adds support for large multi-modal images and tabular data.

Use for: Quick and general-purpose image processing, exploring big image sequences and associated tabular data.



What Fiji looks like



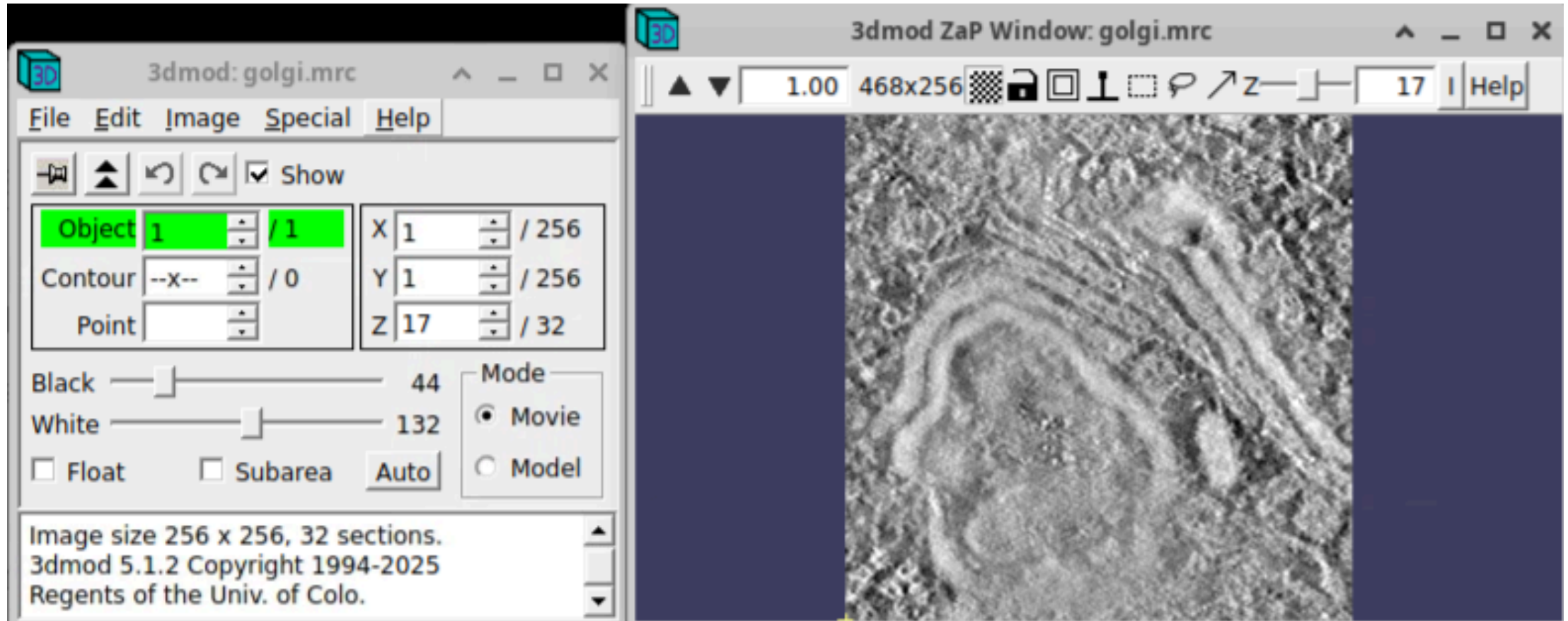
IMOD

IMOD is a bundle of programs for tomographic reconstruction, visualisation and annotation. The main program for viewing and segmenting images is called **3dmod**.

Use for: Aligning and reconstructing tomograms from tilt series (etomo) and 3D visualisation with segmentation (3dmod).

As IMOD comprises many programs, you must run `3dmod` (or `etomo`) in a **Terminal** (type `3dmod` and press enter).

What IMOD looks like (3dmod)



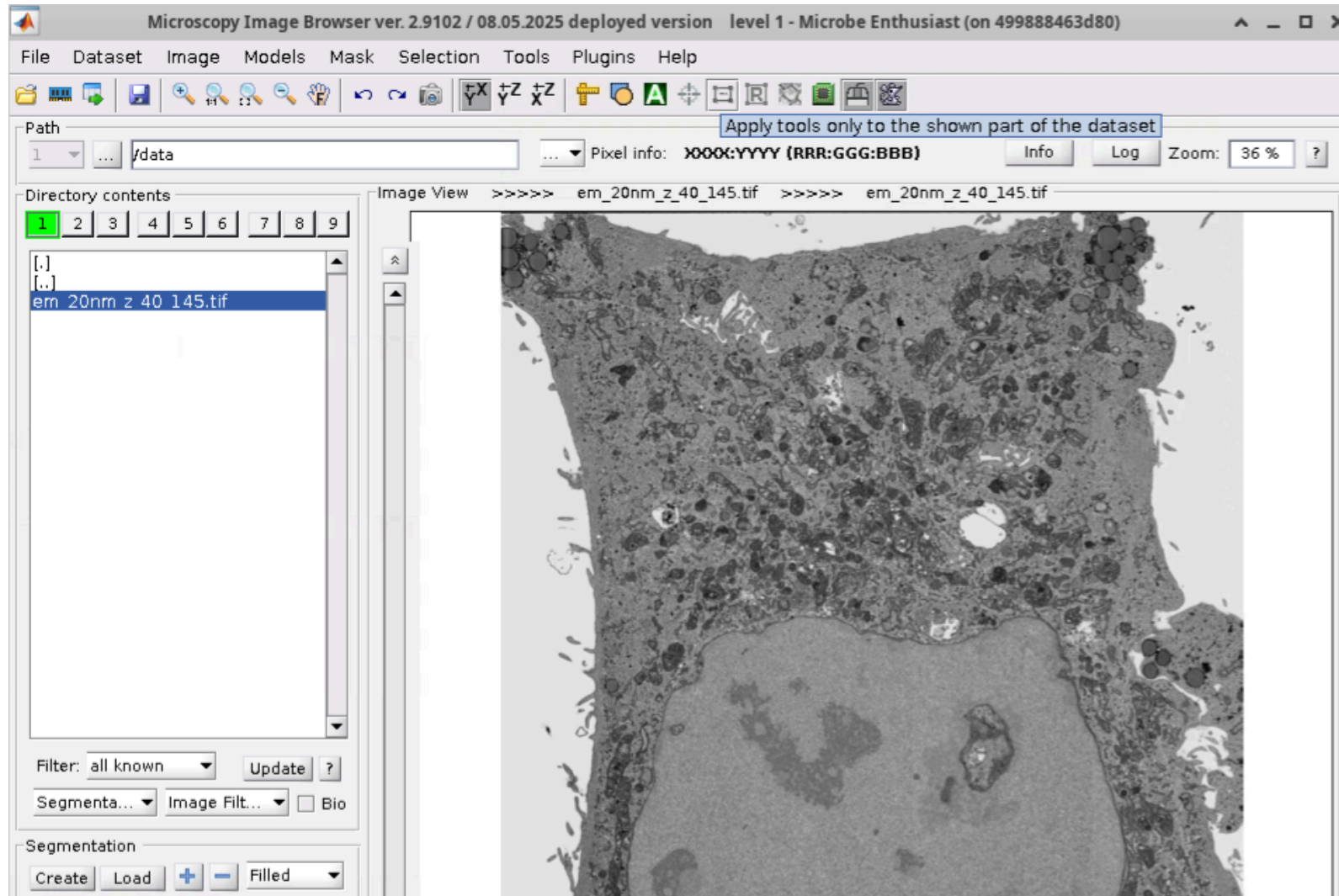
MIB2

Microscopy Image Browser ([MIB](#)) is a user-friendly tool for segmenting and measuring structures in large 3D/4D datasets.

Use for: Annotating regions of interest, extracting quantitative measurements (volumes, surfaces, statistics).



What MIB2 looks like



Wrap-up

CCP-volumeEM containers were created to make image analysis software more accessible

Keep an eye on the [vEM Event List](#) for training and other community events!

Questions? Suggestions? [Create an issue](#) on our documentation repository to let us know.